Medical Chatbot

Final Defense / FDE Project Development
Problem Statement

1. Background

Access to understandable, timely, and trustworthy medical information remains challenging for many people. Online searches often return fragmented, overly technical, outdated, or unverified content. Users may struggle to interpret symptoms, understand common health conditions, or identify when professional medical attention is necessary. These challenges become more pronounced when users prefer conversational explanations or need to ask follow-up questions.

2. Problem Statement

Conventional health information websites largely rely on keyword-based navigation and static articles. They do not always provide a natural, context-aware question-and-answer experience, support different input formats, or clearly distinguish educational information from clinical advice. General-purpose conversational systems can also generate unsupported medical claims, particularly when answers are not grounded in reliable reference material. There is therefore a need for an accessible medical chatbot that combines conversational interaction with evidence retrieval, transparent source references, and appropriate safety guidance.

3. Proposed Solution

This project proposes a web-based medical chatbot that allows users to submit health-related questions through text and, where supported, voice or images. The system will retrieve relevant information from a curated medical knowledge base and may consult current reputable web sources when necessary. An AI-based response component will synthesize accessible explanations and display references when available. The chatbot will communicate uncertainty, avoid presenting itself as a licensed clinician, and advise users to seek professional or emergency care when warranted.

4. Project Objectives

The main objective is to develop and evaluate a user-friendly medical information assistant. Specific objectives are to: (a) implement a responsive conversational web interface; (b) integrate document-based retrieval to ground answers in reference materials; (c) support multimodal input where feasible; (d) present understandable answers with citations and safety disclaimers; and (e) assess response relevance, usability, latency, and safety on a defined set of test questions.

5. Scope and Limitations

The proposed system is intended for educational support and general health information, not diagnosis, prescription, treatment planning, or emergency triage. Image interpretation, if implemented, will be limited to non-diagnostic informational assistance. Responses may be incomplete or incorrect, and the accuracy of external sources may vary. Sensitive user inputs should be handled with appropriate privacy and security controls. Clinical decisions must remain with qualified healthcare professionals.

6. Expected Outcome

The expected deliverable is a functional prototype of a medical chatbot with a responsive interface, retrieval-supported answers, visible source attribution, and clear medical safety messaging. Project evaluation will demonstrate the feasibility of combining conversational AI and a curated knowledge base to improve access to understandable medical information while documenting limitations and areas for future improvement.